Visceral Heterotaxy 13 (HTX13): A Comprehensive Disease Characteristics Report

Disease: Visceral Heterotaxy 13 (HTX13) **OMIM:** #621079 · **MONDO:** MONDO:0976134 · **Gene:** *DAND5* (HGNC:26780, 19p13.13) **Category:** Mendelian, autosomal recessive **Report date:** 2026-10-20

Summary

Visceral Heterotaxy 13 (HTX13; OMIM #621079) is an ultra-rare, autosomal recessive disorder of embryonic left–right (L-R) patterning caused by biallelic (homozygous or compound-heterozygous) loss-of-function (LOF) variants in *DAND5*. *DAND5* (also called *CERL2*, *CER2*, *COCO*, *GREM3*, *DANTE*) encodes a 189-amino-acid secreted cystine-knot protein of the CAN/DAN subfamily of BMP/NODAL antagonists. At the ciliated left–right organizer (LRO) of the early embryo, DAND5 is normally expressed asymmetrically (higher on the right) and acts as a brake on NODAL signaling. When leftward cilia-driven nodal fluid flow degrades DAND5 on the left, NODAL is released to self-amplify, inducing *LEFTY* and then *PITX2* in the left lateral plate mesoderm — the molecular events that specify normal visceral asymmetry (*situs solitus*). Biallelic LOF of DAND5 abolishes this asymmetric brake, resulting in randomized or bilateral NODAL–PITX2 signaling and consequent *situs ambiguus* (heterotaxy).

The clinical phenotype is dominated by complex congenital heart disease (CHD) arising from disrupted cardiac looping and laterality, accompanied by thoraco-abdominal situs anomalies: dextrocardia, atrioventricular septal defects, transposition of the great arteries, anomalous pulmonary venous return, atrial/bronchial isomerism, asplenia or polysplenia, and intestinal malrotation. The HPO gene-to-disease annotation extends the spectrum to include recurrent infections (from asplenia), neurodevelopmental delay, and mild dysmorphism. Disease onset is congenital; inheritance is autosomal recessive, with the disorder thus far reported predominantly in consanguineous Arab-Muslim families.

HTX13 is a very recently delineated entity: the first biallelic LOF case was published in 2022, with the disorder defined through at least three unrelated families (2022–2026) plus in-vitro functional proof that pathogenic DAND5 variants cannot inhibit nodal signaling. Decades of model-organism work (mouse *Dand5*/*Cerl2*, zebrafish *charon*, *Xenopus coco*/*cerl-2*) independently corroborate the mechanism, and gnomAD constraint metrics (pLI = 0.002) are fully consistent with a recessive loss-of-function mechanism rather than haploinsufficiency. No HTX13-specific prevalence, survival, or therapy data exist; management and prognosis are extrapolated from the broader heterotaxy-with-CHD literature (~1/10,000 birth prevalence; ~71%/68% 5-/10-year survival).

Key Findings

Finding 1 — HTX13 is caused by biallelic loss-of-function variants in *DAND5*

OMIM #621079 (Heterotaxy, visceral, 13, autosomal; HTX13) attributes the disorder to **homozygous *DAND5* mutations** (gene MIM 609068) on chromosome 19p13. Three pathogenic alleles have been reported across unrelated families:

- **c.396dupCT** (a 2-bp duplication causing a frameshift; Bolkier et al. 2022)
- **c.197delT / c.197del, p.(Leu66Argfs*22)** (a 1-bp deletion causing a frameshift LOF; Ganapathi et al. 2022)
- A **shared homozygous variant** identified in two unrelated Arab-Muslim probands (Chorin et al. 2026)

Critically, the mechanism has been functionally validated: in a Nodal-dependent luciferase reporter assay, **DAND5 c.197del failed to inhibit Nodal signaling, whereas the wild-type construct did**, demonstrating that the frameshift produces a complete loss of function. The convergence of independent human families, a reproducible biochemical defect, and strong model-organism support establishes *DAND5* as the causal gene for HTX13.

> *"ES revealed a shared homozygous variant in the Dan Domain Family Member 5"* — PMID: 42650045 (Chorin et al. 2026)
>
> *"A novel biallelic loss-of-function variant in..."* — PMID: 36316122 (Ganapathi et al. 2022), reporting the c.197delT allele.

Finding 2 — DAND5/Cerl2 is an asymmetrically expressed NODAL antagonist at the left–right organizer

In the mouse node, **Nodal and Cerl2 (the mouse ortholog of DAND5) are expressed asymmetrically**: Nodal is higher on the left, while Cerl2 — a Nodal inhibitor — is higher on the right. Leftward nodal fluid flow reduces Cerl2 on the left side, releasing NODAL to self-amplify and induce *Lefty* and then *Pitx2* in the left lateral plate mesoderm. The *inv* mutant demonstrates the causal logic directly: a localized reduction of Cerl2 up-regulates Nodal and reverses L-R asymmetry (Oki et al. 2009). More broadly, **DAND5, PKD1L1, MMP21, and CIROP form a genetic module** that has specifically evolved in vertebrate species harboring motile cilia at the LRO (Szenker-Ravi et al. 2025), placing DAND5 at the heart of a conserved laterality-determining mechanism.

> *"the expression of Nodal and Cerl2 (Dand5), which encodes an inhibitor of Nodal, is asymmetric, with the level of Nodal expression being higher on the left side and that of Cerl2 expression higher on the right"* — PMID: 19906859 (Oki et al. 2009)
>
> *"Four genes-DAND5, PKD1L1, MMP21, and CIROP-form a genetic module that has specifically evolved in vertebrate species that harbor motile cilia in their left-right organizer (LRO)"* — PMID: 39753129 (Szenker-Ravi et al. 2025)

Finding 3 — *Dand5/Cerl2* knockout mice model HTX13: laterality randomization plus intrinsic cardiac defects

Cerl2(−/−) mice show significant neonatal mortality within the first day after birth, **largely attributable to cardiac defects associated with randomization of the left–right axis**. Importantly, and independent of laterality, Cerl2-null neonates also develop massive **left-ventricular myocardial wall enlargement (hyperplasia)** driven by a high cardiomyocyte mitotic index with increased *Ccnd1* expression; echocardiography reveals left-ventricular systolic dysfunction incompatible with a long lifespan (Araújo et al. 2014). Cerl2 is normally expressed more highly in the left ventricle at E13, indicating a regionalized role in left-ventricular muscle formation that is distinct from its role in axis specification. This dual phenotype — laterality randomization plus intrinsic myocardial dysfunction — makes the Cerl2 knockout an informative model for the cardiac lethality seen in human heterotaxy.

> *"revealed a significant mortality during the first day after birth, mostly due to cardiac defects apparently associated with randomization of the left-right axis"* — PMID: 25033293 (Araújo et al. 2014)

Finding 4 — DAND5 is a 189-aa secreted cystine-knot NODAL/BMP antagonist; gnomAD constraint confirms a recessive mechanism

DAND5 (HGNC:26780; NCBI Gene 199699; UniProt Q8N907; Ensembl ENSG00000179284; cytoband 19p13.13) encodes a **189-residue secreted protein** with an N-terminal signal peptide (aa 1–22) and a C-terminal **cystine-knot (CTCK) domain (aa 101–186)** stabilized by four disulfide bonds (Cys101–148, 115–162, 125–183, 129–185); it is N-glycosylated at N38. DAND5 belongs to the **CAN/DAN subfamily of BMP antagonists**, binds NODAL to inhibit left–right patterning, and additionally antagonizes BMP/TGF-β signaling and exhibits anti-angiogenic activity.

Population constraint metrics from gnomAD are decisive for the inheritance model: **pLI = 0.002** and **LOEUF/oe_lof = 0.87** (observed LOF = 8 vs. expected = 9.24). DAND5 is therefore **not intolerant to heterozygous loss of function** — consistent with the clinical observation that heterozygous carriers are unaffected and disease requires biallelic LOF. The two reported pathogenic frameshifts — **c.197delT** (~codon 66, upstream of the CTCK domain) and **c.396dupCT** (within the CTCK domain) — are both predicted to **abolish the cystine-knot fold**, consistent with complete loss of function.

Attribute	Value
Gene symbol	*DAND5* (aliases *CERL2*, *CER2*, *COCO*, *GREM3*, *DANTE*)
HGNC	HGNC:26780
NCBI Gene	199699
UniProt	Q8N907
Ensembl	ENSG00000179284
Cytoband	19p13.13
Protein length	189 aa
Key domains	Signal peptide (1–22); CTCK cystine-knot (101–186)
gnomAD pLI	0.002 (LOF-tolerant → recessive mechanism)
gnomAD LOEUF	0.87

Finding 5 — HTX13 phenotype spectrum (HPO) extends beyond cardiac/visceral situs to asplenia/polysplenia, neurodevelopmental delay, and dysmorphism

The HPO/JAX gene-to-disease annotation for *DAND5* (OMIM:621079) comprises **29 phenotype terms**, organized below by system. Onset is **Congenital (HP:0003577)** and inheritance is **Autosomal recessive (HP:0000007)**.

Category	HPO phenotypes (term IDs)
Situs / laterality	Isomerism (HP:0031853), Right atrial isomerism (HP:0011536), Bronchial isomerism (HP:0031564), Dextrocardia (HP:0001651)
Cardiac	Complete atrioventricular canal defect (HP:0001674), Transposition of the great arteries (HP:0001669), Valvular/pulmonic stenosis (HP:0034350 / HP:0001642), Patent ductus arteriosus (HP:0001643), Infracardiac TAPVC (HP:0011721), Partial anomalous pulmonary venous return (HP:0010773), Cardiomyopathy (HP:0001638), Cardiomegaly (HP:0001640), Congestive heart failure (HP:0001635), Pulmonary arterial hypertension (HP:0002092)
Spleen / immune	Asplenia (HP:0001746), Polysplenia (HP:0001748), Recurrent infections (HP:0002719)
Gastrointestinal	Intestinal malrotation (HP:0002566), Nasogastric tube feeding (HP:0040288)
Neurodevelopmental	Global developmental delay (HP:0001263), Delayed speech/language (HP:0000750), Delayed fine motor development (HP:0010862), Dyskinesia (HP:0100660)
Dysmorphism	Hypertelorism (HP:0000316), Epicanthus (HP:0000286), Coarse facial features (HP:0000280)

The breadth of this annotation reflects the pleiotropic downstream consequences of disrupted L-R patterning (cardiac, splenic, GI situs) plus secondary complications (recurrent infection from asplenia; feeding difficulty), and some features (neurodevelopmental delay, dysmorphism) that are reported but whose causal relationship to DAND5 loss is less firmly established given the small number of cases.

Finding 6 — ClinVar documents two heterotaxy-associated *DAND5* frameshifts; missense variants are VUS, and large 19p13 CNVs cause other syndromes

On transcript **NM_152654.3**, ClinVar records the two HTX13 variants:

- **c.197del, p.(Leu66fs)** — condition: Heterotaxy; currently classified **Uncertain significance** (single-submitter, reflecting the rarity of the disorder despite functional evidence)
- **c.396_397dup, p.(Tyr133fs)** — condition: Heterotaxy; classified **Pathogenic/Likely pathogenic**

By contrast, **all ~20 recorded DAND5 single-nucleotide (missense) variants** (e.g., p.Arg152Cys, p.Arg112Trp, p.Gly80Glu) are classified **Benign/Likely benign or Uncertain significance**, with no established disease link — underscoring that the established disease mechanism is frameshift/truncating LOF, not missense. Separately, numerous **large 19p13 copy-number variants** (deletions/duplications spanning *DAND5*) in ClinVar are annotated to **contiguous-gene conditions** (e.g., episodic ataxia type 2 at the *CACNA1A* region; developmental and epileptic encephalopathy 42) rather than to isolated HTX13 — an important caveat for interpreting CNVs overlapping this locus.

Finding 7 — Precise variant nomenclature, functional proof, and the full cross-species ortholog set

Ganapathi et al. 2022 (PMID: 36316122) describe the **second unrelated biallelic HTX13 individual**, homozygous for **NM_152654.2:c.197del, p.(Leu66ArgfsTer22)**; an in-vitro assay showed this variant cannot inhibit nodal signaling versus wild-type, strengthening the evidence for autosomal recessive DAND5 heterotaxy. The **first biallelic LOF case** was reported by Bolkier et al. 2022 (PMID: 34215651), within a cohort in which WES achieved a molecular diagnosis in 56.7% of laterality/CHD probands.

DAND5 orthologs (HomoloGene 51883 and model-organism databases):

Species	Gene	NCBI Gene ID
Human	*DAND5*	199699
Mouse	*Dand5* / *Cerl2*	23863
Rat	*Dand5*	685719
Dog	*Dand5*	484921
Rhesus macaque	*Dand5*	717076
Chimpanzee	*Dand5*	747680
Zebrafish	*dand5* / *charon*	791454
Xenopus tropicalis	*dand5* / *cerl-2* / *coco*	733727

> *"we describe a second unrelated individual with heterotaxy syndrome and a homozygous frameshift variant in DAND5 (NM_152654.2:c.197del..."* — PMID: 36316122

> *"the DAND5 c.197del variant is unable to inhibit nodal signaling when compared with the wild-type expression construct"* — PMID: 36316122

Finding 8 — HTX13 is an ultra-rare, recently defined (2022–2026) autosomal recessive laterality disorder with limited but convergent evidence

Gene–disease validity rests on **≥3 unrelated biallelic DAND5 families** (Bolkier 2022 [PMID: 34215651]; Ganapathi 2022 [PMID: 36316122]; Chorin 2026 [PMID: 42650045]), **in-vitro functional proof** of Nodal-inhibition loss, **decades of model-organism data** (mouse *Cerl2*/*Dand5*, zebrafish *charon*, *Xenopus coco*), and **gnomAD constraint consistent with recessive LOF** (pLI = 0.002). No HTX13-specific prevalence, incidence, sex ratio, survival statistics, pharmacotherapy, or disease-specific therapy exist. Prognosis and management are extrapolated from the general heterotaxy-with-CHD literature, where **~1/10,000 birth prevalence** and **5-/10-year survival of ~71%/68%** are reported (context: PMID: 42185088). Management is supportive/surgical (staged CHD repair, Ladd procedure for malrotation, asplenia infection prophylaxis); prevention consists of genetic counseling, carrier/cascade screening, and prenatal detection.

Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic LOF variant in *DAND5*** (e.g., c.197del p.Leu66Argfs*22 or c.396dupCT)* → *leads to** a truncated DAND5 protein lacking/disrupting the C-terminal cystine-knot (CTCK) domain.
2. Loss of the cystine-knot fold → **results in** a secreted protein unable to bind and inhibit NODAL (demonstrated: the c.197del variant fails to inhibit Nodal signaling in a luciferase reporter assay).
3. Absence of functional DAND5 at the ciliated left–right organizer (LRO) → **removes** the normal asymmetric "brake" on NODAL (normally higher-right DAND5 restricting NODAL to the left).
4. Loss of the asymmetric brake → **causes** randomized or bilateral NODAL self-amplification instead of normal left-restricted signaling. *(This is the branch point: the readout is stochastic because the symmetry-breaking cue is lost rather than reversed.)*

5. Dysregulated NODAL → **drives** abnormal/bilateral *LEFTY* and *PITX2* induction in the lateral plate mesoderm (inferred in human from the conserved vertebrate cascade; demonstrated in mouse/frog/fish models).
6. Loss of asymmetric *PITX2* → **results in** abnormal cardiac looping and failure of normal visceral situs specification.
7. Abnormal looping/situs → **manifests as** *situs ambiguus* (heterotaxy): dextrocardia, complex CHD (AVSD, TGA, anomalous pulmonary venous return), atrial/bronchial isomerism, asplenia or polysplenia, and intestinal malrotation.
8. In parallel (laterality-independent branch, demonstrated in mouse): loss of DAND5/Cerl2 → **causes** left-ventricular cardiomyocyte hyperplasia (↑ Ccnd1, ↑ mitotic index) → **leads to** systolic dysfunction — contributing to cardiac morbidity beyond the situs defect itself.
9. Downstream clinical complications → asplenia **leads to** susceptibility to encapsulated-organism infection; malrotation **leads to** risk of midgut volvulus; complex CHD **leads to** heart failure and perinatal/infant mortality.

Schematic

```
DAND5 biallelic LOF
        |
        v
Truncated protein, disrupted cystine-knot (CTCK) domain
        |
        v
Loss of NODAL-antagonist function  --(functionally proven: no Nodal inhibition)
        |
        v
LRO: asymmetric "brake" on NODAL removed
        |
        +---------------- Laterality branch ----------------+
        v                                                    v
Randomized / bilateral NODAL             (mouse) LV cardiomyocyte hyperplasia
        |                                        up Ccnd1, up mitotic index
        v                                                    |
Abnormal LEFTY / PITX2                                       v
        |                                        LV systolic dysfunction
        v
Abnormal cardiac looping + visceral situs failure
        |
        v
SITUS AMBIGUUS (HETEROTAXY):
  dextrocardia . complex CHD (AVSD, TGA, anomalous pulm. venous return)
  atrial/bronchial isomerism . asplenia/polysplenia . intestinal malrotation
        |
        v
Complications: infection (asplenia) . volvulus (malrotation) . heart failure
```

Upstream vs. downstream, cell types, and processes

- **Upstream (initiating):** The *DAND5* LOF mutation and the resulting loss of NODAL antagonism at the LRO are the most upstream events. The relevant cell types are the **ciliated cells of the left–right organizer** (node/gastrocoel roof plate), bearing both motile and immotile (sensory) cilia.
- **Midstream:** Dysregulated NODAL–LEFTY–PITX2 signaling in the **lateral plate mesoderm**.
- **Downstream:** Abnormal morphogenesis of **cardiomyocytes** (looping heart tube, LV myocardium), **splenic mesenchyme**, and **gut endoderm/mesenchyme**, producing the organ-level phenotype.

Suggested ontology terms: - *Biological process (GO):* determination of left/right symmetry (GO:0007368); nodal signaling pathway (GO:0038092); BMP signaling pathway (GO:0030509); heart looping (GO:0001947); cilium movement (GO:0003341). - *Molecular function (GO):* BMP binding (GO:0036122); growth factor activity / cytokine antagonist activity. - *Cellular component (GO):* extracellular space (GO:0005615) — DAND5 is secreted. - *Cell types (CL):* ciliated cell (CL:0000064); cardiac muscle cell / cardiomyocyte (CL:0000746); epithelial cell of the left–right organizer. - *Anatomy (UBERON):* embryonic node / left–right organizer; lateral plate mesoderm (UBERON:0003081); heart (UBERON:0000948); spleen (UBERON:0002106); intestine (UBERON:0000160). - *Chemical (CHEBI):* no disease-specific metabolite; NODAL/BMP are proteins, not CHEBI entities.

Section-by-Section Characteristics

1. Disease Information

HTX13 is a Mendelian disorder of visceral left–right asymmetry (situs ambiguus / heterotaxy) with complex congenital heart disease. **Identifiers:** OMIM #621079; MONDO:0976134; gene *DAND5* (MIM 609068). ICD-10 Q89.3 (situs inversus) / Q20–Q24 (congenital heart malformations) are the closest available codes; there is no HTX13-specific ICD code. **Synonyms:** Heterotaxy, visceral, 13, autosomal (HTX13); DAND5-related heterotaxy; *DAND5* aliases include CERL2, CER2, COCO, GREM3, DANTE. Information is derived from **aggregated disease-level resources** (OMIM, HPO, ClinVar) and a small number of **individual patient case reports**.

2. Etiology

Causal factor: genetic — biallelic LOF variants in *DAND5*. **Genetic risk factors:** the causal variants themselves (frameshifts); **consanguinity** is a major risk factor, as all reported families are consanguineous (Arab-Muslim descent). No environmental or infectious cause is established for HTX13 specifically. In the broader heterotaxy/laterality literature, **pregestational/maternal diabetes** is a recognized environmental risk factor for laterality spectrum disorders (PMID: 41974186), and some reports describe a possible rise in fetal laterality disorders around the COVID-19 pandemic (PMID: 42110452) — but neither has been linked to DAND5-specific disease. **Protective factors:** none identified. **Gene–environment interactions:** not characterized for HTX13.

3. Phenotypes

See Finding 5 for the complete 29-term HPO list. The dominant, highest-frequency phenotypes are **cardiac** (complex CHD, dextrocardia, AVSD, TGA, anomalous pulmonary venous return) and **situs/laterality** (isomerism, asplenia/polysplenia, intestinal malrotation). Severity is **variable but often severe**, driven by the cardiac lesion. Onset is **congenital/neonatal**; the course is determined by CHD severity. Quality-of-life impact is substantial in affected infants (feeding difficulty requiring NG tube; surgical burden; infection risk from asplenia), though no HTX13-specific QoL instrument data exist.

4. Genetic/Molecular Information

Causal gene: *DAND5* (see Finding 4 for full identifiers). **Pathogenic variants:** c.197del p.(Leu66Argfs*22) and c.396_397dup p.(Tyr133fs), both frameshift/truncating, both predicted/proven LOF. *Variant classification (ACMG/ClinVar): c.396_397dup = Pathogenic/Likely pathogenic; c.197del = Uncertain significance despite functional evidence (single submitter). Allele frequency: both are ultra-rare/absent in gnomAD; recessive disease requires biallelic LOF (gnomAD pLI = 0.002 confirms LOF-tolerance in heterozygotes). Origin: germline. Functional consequence: loss of function (abolished NODAL inhibition). Missense variants across DAND5 are benign/VUS. Modifier genes/epigenetics: not characterized. Chromosomal abnormalities: large 19p13 CNVs spanning* DAND5* are associated with contiguous-gene syndromes, not isolated HTX13 (Finding 6).

5. Environmental Information

No DAND5/HTX13-specific environmental, lifestyle, or infectious factor is established. Context from the broader laterality field implicates maternal pregestational diabetes as a risk factor for laterality spectrum disorders (PMID: 41974186).

6. Mechanism / Pathophysiology

Fully detailed in the **Mechanistic Model** section above: biallelic DAND5 LOF → loss of NODAL antagonism at the LRO → randomized/bilateral NODAL–LEFTY–PITX2 signaling → abnormal cardiac looping and visceral situs → heterotaxy, with a parallel laterality-independent myocardial-hyperplasia branch demonstrated in mouse. Key pathway: **NODAL/TGF-β superfamily signaling**; antagonized BMP signaling; cilia-driven nodal flow.

7. Anatomical Structures Affected

Primary organs: heart (looping, septation, great-vessel and pulmonary-venous connections), spleen (asplenia/polysplenia), intestine (malrotation). **Body systems:** cardiovascular, digestive, immune/lymphatic (spleen), respiratory (bronchial isomerism). **Tissue/cell level:** cardiomyocytes (CL:0000746); ciliated LRO cells (CL:0000064); splenic and gut mesenchyme. **Subcellular:** extracellular space (secreted DAND5); cilium/basal body at the LRO. **Localization:** thoraco-abdominal; lateralization is the defining abnormality — **abnormal/randomized laterality (isomerism, situs ambiguus)** rather than consistent unilateral/bilateral.

8. Temporal Development

Onset: congenital — the lesion acts during gastrulation/early organogenesis (symmetry-breaking at the LRO is an early-embryonic event; the critical window is the node stage). **Progression:** the structural defects are fixed at birth; clinical course is driven by CHD and its surgical palliation. **Disease course:** chronic/lifelong for survivors; mortality concentrated in the neonatal/infant period. **Critical period for intervention:** prenatal detection and perinatal/surgical planning.

9. Inheritance and Population

Inheritance: autosomal recessive (HP:0000007). **Penetrance:** appears high in biallelic individuals, but the small case number precludes precise estimates. **Expressivity:** variable (range and severity of cardiac/visceral anomalies differ between families). **Consanguinity:** central — all reported families are consanguineous. **Carrier frequency:** not established; DAND5 LOF alleles are ultra-rare in gnomAD. **Founder effect:** a shared homozygous variant in two unrelated Arab-Muslim probands (Chorin 2026) raises the possibility of a regional founder allele. **Epidemiology:** no HTX13-specific prevalence; heterotaxy overall ~1/10,000 births. **Sex ratio / age distribution:** not established for HTX13.

10. Diagnostics

Imaging: fetal echocardiography and prenatal ultrasound are the mainstay for detecting situs/cardiac anomalies (heterotaxy has among the highest prenatal detection rates, e.g., 91% in one Japanese cohort, PMID: 42324491); postnatal echocardiography, CT/MRI, abdominal ultrasound (spleen, situs). **Laboratory:** peripheral smear for Howell–Jolly bodies (functional asplenia). **Genetic testing:** WES/WGS is the diagnostic modality that defined HTX13 (WES diagnosed 56.7% of a laterality/CHD cohort, PMID: 34215651); targeted *DAND5* sequencing or heterotaxy/

CHD gene panels; chromosomal microarray to exclude contiguous-gene CNVs. **Clinical criteria:** diagnosis based on demonstration of situs ambiguus with complex CHD plus biallelic *DAND5* LOF. **Differential diagnosis:** other heterotaxy genes (PKD1L1/HTX8, ANKS3, CCDC11, MMP21, ZIC3/HTX1, NODAL, CFC1, etc.) and primary ciliary dyskinesia.

11. Outcome/Prognosis

No HTX13-specific survival data. Extrapolating from heterotaxy-with-CHD cohorts: median **5-/10-year survival ~71%/68%**; **complete heart block and pulmonary venous obstruction** are key predictors of mortality (PMID: 42185088). Asplenia-syndrome single-ventricle palliation shows ~75.7% 1-year and ~53.7% 20-year survival, worse with TAPVC/pulmonary atresia/AV-valve regurgitation (PMID: 41614290). **Morbidity:** surgical burden, infection risk (asplenia), feeding difficulty, developmental delay. **Prognostic factors:** cardiac morphology severity, RVOT obstruction, heart block, pulmonary venous obstruction.

12. Treatment

No disease-specific or gene-targeted therapy exists. Management is **supportive and surgical**: staged congenital heart surgery (often single-ventricle palliation/Fontan pathway for complex lesions); **Ladd procedure** for intestinal malrotation; **antibiotic prophylaxis and vaccination against encapsulated organisms** for asplenia; nutritional support (NG/feeding support). Pharmacotherapy is symptomatic (heart-failure management, arrhythmia/heart-block management including pacing). **Suggested NCIT terms:** Congenital Heart Disease Surgery; Fontan Procedure; Antibiotic Prophylaxis; Vaccination. No pharmacogenomic, gene, cell, or RNA therapy is available or in trials for HTX13 specifically.

13. Prevention

Primary prevention: not applicable (congenital genetic disorder) beyond avoidance of modifiable maternal risk factors (e.g., glycemic control in pregestational diabetes, by extrapolation). **Secondary prevention / early detection:** prenatal ultrasound and fetal echocardiography for early diagnosis and delivery planning. **Genetic prevention: genetic counseling** for consanguineous families, **carrier/cascade screening** of relatives, **prenatal diagnosis** and **preimplantation genetic testing** for families with a known biallelic genotype. **Tertiary prevention:** asplenia infection prophylaxis; surveillance for and timely repair of malrotation to prevent volvulus.

14. Other Species / Natural Disease

DAND5 is highly conserved across vertebrates (orthologs in mouse, rat, dog, rhesus, chimp, zebrafish, *Xenopus* — Finding 7). The mechanism is evolutionarily conserved: DAND5/PKD1L1/MMP21/CIROP form a vertebrate LRO module tied to motile cilia (PMID: 39753129). No naturally occurring companion-animal/wildlife DAND5 disease is catalogued in OMIA for this entity; the disorder is essentially defined through experimental models. No zoonotic potential (non-infectious disorder).

15. Model Organisms

- **Mouse** (*Dand5*/*Cerl2*): knockout recapitulates laterality randomization + neonatal cardiac lethality, plus laterality-independent LV hyperplasia/systolic dysfunction (Araújo 2014, PMID: 25033293); *inv* mutant shows Cerl2 reduction reverses asymmetry (Oki 2009, PMID: 19906859). **Strong phenotype recapitulation** of the laterality/cardiac axis.
- **Zebrafish** (*dand5*/*charon*): laterality model; *dand5* is a standard L-R marker readout.
- **Xenopus** (*dand5*/*coco*/*cerl-2*): laterality model; *dand5* expression is a canonical early L-R asymmetry marker (used in CACNA1G, CCDC57 studies — PMID: 40008628, PMID: 41758249).
- **In vitro:** Nodal-dependent luciferase reporter assay provided direct functional proof that patient variants cannot inhibit Nodal (Ganapathi 2022).
- **Limitations:** species differences in cilia dependence of L-R patterning; models capture the laterality/cardiac mechanism well but not the full human neurodevelopmental/dysmorphic spectrum. **Resources:** MGI, ZFIN, Xenbase, IMPC.

Evidence Base

PMID	Study (abbrev.)	Type	How it supports/challenges findings
34215651	Bolkier et al. 2022 — WES in 30 Arab-Muslim laterality/CHD probands	Human clinical	First biallelic DAND5 LOF case (c.396dupCT); WES diagnosed 56.7% of cohort — supports Findings 1, 7, 8
36316122	Ganapathi et al. 2022 — second DAND5 individual	Human clinical + in vitro	Exact variant c.197del p.Leu66Argfs*22; functional proof of lost Nodal inhibition — supports Findings 1, 7
42650045	Chorin et al. 2026 — further support	Human clinical	Shared homozygous DAND5 variant in two unrelated probands (possible founder) — supports Findings 1, 8
19906859	Oki et al. 2009 — reversal of L-R asymmetry	Mouse model	Establishes asymmetric Nodal/Cerl2 expression and causal role of Cerl2 reduction — supports Finding 2
25033293	Araújo et al. 2014 — Cerl2 inactivation	Mouse model	Knockout models laterality-associated cardiac lethality + LV hyperplasia — supports Finding 3
39753129	Szenker-Ravi et al. 2025 — CIROZ/LRO module	Computational/ comparative	Places DAND5 in conserved vertebrate LRO genetic module — supports Finding 2
38884745	Molecular Pathways and Animal Models of Situs Defects (review)	Review	Frames NODAL-left/BMP-right paradigm and LRO cilia biology — supports mechanistic model
42185088	Risk Factors for Adverse Outcome in Heterotaxy+CHD fetuses	Human clinical	Provides 5-/10-yr survival (71%/ 68%) and prognostic factors — context for prognosis (Finding 8)
41614290	Long-term outcomes after single-ventricle palliation, asplenia	Human clinical	Asplenia-syndrome survival and surgical risk factors — context for prognosis/treatment

PMID	Study (abbrev.)	Type	How it supports/challenges findings
42324491	Prenatal detection of critical CHD, Japan	Human clinical	Heterotaxy prenatal detection ~91% — context for diagnostics
41974186	Laterality spectrum disorders + pregestational diabetes	Human clinical	Maternal diabetes association — context for etiology/environment
40008628	CACNA1G in Xenopus L-R patterning	Frog model	Uses dand5 as early L-R marker — supports DAND5's upstream role
41758249	Ccdc57 regulates cilia/L-R patterning	Frog model + human	dand5 misexpression as readout of LRO dysfunction — supports mechanism

Note on citation integrity: Two citation snippets for

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are flagged as "mismatch" in the knowledge state (the exact verbatim text may differ slightly from the stored abstract). The substantive claims — that this paper reports a biallelic DAND5 c.197del in a second heterotaxy patient and demonstrated loss of Nodal inhibition — are consistent with the paper's content as recorded; readers should verify exact quotations against the primary source.

Limitations and Knowledge Gaps

1. **Very small case base.** HTX13 rests on ~3 unrelated families (2022–2026). Penetrance, expressivity, sex ratio, and the full phenotype spectrum are poorly defined. The neurodevelopmental and dysmorphic HPO annotations in particular may reflect single-case observations or coincidental features rather than core DAND5 consequences.
2. **Variant classification lag.** Despite functional proof, c.197del remains a single-submitter VUS in ClinVar, illustrating how rare-disease evidence propagates slowly into variant databases.
3. **No HTX13-specific epidemiology, natural history, survival, or treatment data.** All prognosis/management statements are extrapolated from general heterotaxy-with-CHD

cohorts, which include many non-DAND5 etiologies and may not reflect DAND5-specific outcomes.

4. **Mechanistic inference in humans.** The NODAL–LEFTY–PITX2 downstream cascade and the LV-hyperplasia branch are demonstrated in model organisms; their precise operation in human DAND5 patients is inferred, not directly shown.
5. **Ascertainment bias toward consanguineous populations.** Reported families are Arab-Muslim/consanguineous; the allele spectrum and frequency in other populations are unknown.
6. **No genotype–phenotype map.** With only truncating alleles reported as pathogenic, the clinical consequences of hypomorphic/missense DAND5 variants (currently VUS/benign) are unresolved.

Proposed Follow-up Experiments / Actions

1. **Aggregate cases via GeneMatcher / Matchmaker Exchange** to expand the HTX13 cohort, enabling estimation of penetrance, expressivity, sex ratio, and a validated phenotype spectrum (clarifying whether neurodevelopmental/dysmorphic features are core).
2. **Upgrade ClinVar classification** of c.197del by submitting the published functional (Nodal-inhibition) evidence under ACMG PS3, and encourage multi-submitter review.
3. **Founder-allele analysis** in Arab-Muslim populations: genotype the shared Chorin 2026 variant, estimate carrier frequency in regional/local cohorts, and assess a founder haplotype — informing carrier-screening programs.
4. **Patient-derived iPSC → cardiac organoid / LRO-like models** to test whether human DAND5 LOF reproduces the mouse LV-hyperplasia branch and to dissect NODAL–PITX2 dysregulation in a human context.
5. **Functional assays for DAND5 missense VUS** (Nodal-inhibition reporter) to build a genotype–function map and resolve whether any missense alleles are hypomorphic.
6. **Prospective natural-history registry** for molecularly confirmed DAND5 heterotaxy to generate disease-specific survival, surgical-outcome, and quality-of-life data rather than relying on mixed-etiology cohorts.
7. **Prenatal/perinatal pathway**: integrate DAND5 testing into heterotaxy fetal-cardiology workups given the high prenatal detection rate of heterotaxy, enabling earlier counseling and delivery planning.

Consensus Answer

Visceral Heterotaxy 13 (HTX13; OMIM #621079, MONDO:0976134) is an ultra-rare autosomal recessive disorder of embryonic left–right patterning caused by biallelic loss-of-function variants in *DAND5* (HGNC:26780, 19p13.13; e.g., c.197del p.Leu66Argfs*22 and c.396_397dup p.Tyr133fs), which encodes a secreted cystine-knot NODAL antagonist at the ciliated left–right organizer. Loss of DAND5 removes the asymmetric brake on NODAL, causing randomized/bilateral NODAL–PITX2 signaling and consequent situs ambiguus with complex congenital heart disease (dextrocardia, AVSD, TGA, anomalous pulmonary venous return), isomerism, asplenia/polysplenia, and intestinal malrotation; management is supportive/surgical with no disease-specific therapy, and mouse* Dand5/Cerl2*, zebrafish* charon*, and* Xenopus coco* models recapitulate the laterality-plus-cardiac phenotype.